



December 2025

(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

13. Source segregation of waste means separating waste at _____.
 A) Landfill B) Incinerator
 C) Point of generation D) Transfer station
14. Windrow composting involves _____.
 A) Closed reactors B) Incineration
 C) Landfilling D) Long piles turned periodically
15. C/N ratio desirable for composting is around _____.
 A) 10:1 B) 20:1
 C) 50:1 D) 100:1
16. Hazardous wastes are characterized by _____.
 A) Ignitability, corrosivity, reactivity, toxicity B) Color only
 C) Odor only D) Moisture
17. Phytostabilization achieves _____.
 A) Metal immobilization in soil B) Gas release
 C) Water filtration only D) Pathogen kill
18. Biofilms are defined as _____.
 A) Single free cells B) Viruses in suspension
 C) Sterile layers D) Communities of microorganisms attached to a surface
19. Lignocellulosic biomass requires _____ before ethanol fermentation.
 A) Deep freezing B) Painting
 C) Pretreatment & enzymatic hydrolysis D) No treatment
20. Biotechnology assists biodiversity conservation by _____.
 A) Deforestation B) Species extinction
 C) Creating monocultures only D) Genetic characterization and germplasm storage

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) Summarize any two biotechnological means of pollution detection.	8	L*2	1	1,5
b) What are polychlorinated biphenyls? What are their properties? How are they biodegraded?	8	L2	2	2
2. a) Is experimental measurement of bioaccumulation possible? How? Discuss.	8	L2	1	1
b) Phenolic compounds can undergo mineralization upon microbial action. Justify.	8	L3	2	2
3. a) What are the different factors affecting bioaccumulation? How do they influence bioaccumulation rate?	8	L2	1	1
b) Why are synthetic detergents preferred? Are they biodegradable? Discuss.	8	L2	2	2

Unit – II

4. a) Discuss about physical, chemical, and biological characteristics of wastewater.	8	L2	3	1
b) Outline biomedical waste management practices.	8	L2	4	1
5. a) Elaborate on biological phosphorus removal.	8	L2	3	1,7
b) Explain the process of anaerobic digestion of solid wastes.	8	L2	4	1
6. a) Describe skimming tanks with reference to preliminary wastewater treatment.	8	L2	3	1
b) Sanitary landfills are regarded as engineered landfills. Justify.	8	L3	4	2,7

Unit – III

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|----|----|--|---|----|---|-------|
| 7. | a) | Give a detailed account on phytoremediation. | 8 | L2 | 5 | 1 |
| | b) | Can bacteria act as biopesticides? Discuss with the help of any one example. | 8 | L3 | 5 | 2,7 |
| 8. | a) | Outline the process of second-generation bioethanol production. | 8 | L2 | 5 | 2,6,7 |
| | b) | Discuss different aspects of bioleaching. | 8 | L2 | 5 | 2,6,7 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
